CS23E01 / EMBEDDED SYSTEMS AND INTERNET OF THINGS

- Session: July-Nov 2026 (B.E. CSE – VII sem)

Activities for Lab-8:

Topics: Interfacing with an LCD display (Virtual Hardware) on MCU 8051 IDE, RASPBERRY PI pin diagram, hands-on with actual kit. Using any available Raspberry Pi Simulators.

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1. Interface with an LCD/7-segment display

2. using 7-segment LCD display (virtual hardware) with delay for displaying numbers from 0 to 9 in a loop.

3. implement a simple digital counter that counts the number of times a button (high on P1) is pressed and displays the count on any display panel).

4. Some Raspberry Pi Simulators online:
   https://create.withcode.uk/python/A5,

   Basic Pico boards:

   WokWi: https://wokwi.com/pi-pico

   Velxio: https://velxio.dev/editor

5. Understand the pin diagram, Pi kit, board, sensors, working of Raspberry Pi (refer to the material shared)

6. Demonstrate LED blinking.

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Some code snippet – LED Blinking

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```
import RPi.GPIO as GPIO
from time import sleep
GPIO.setwarnings(False) # Ignore warning for now
GPIO.setmode(GPIO.BOARD) # Use physical pin numbering
GPIO.setup(8, GPIO.OUT, initial=GPIO.LOW) # Set pin 8 to be an output pin and set initial value to low (off)
while True:
 GPIO.output(8, GPIO.HIGH)
 sleep(1)
 GPIO.output(8, GPIO.LOW)
 sleep(1)
```

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